SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

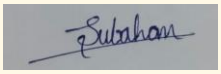
CO1 ASSESSMENT TOOL 2 – Agile Sprint Planning & User Story Workshop

Course Code:	CSA10 / CSA1063	**Course Title:**	Software Engineering
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – **Agile Sprint Planning & User Story Workshop**
Weightage:	20%	**Total Marks:**	25
CO1 Statement:	*Apply advanced methodologies and agile project management techniques to manage software projects effectively*		
Student Name:	T. Subahan	**Reg. No.:**	192511121
Date:	12/10/26	**Duration:**	60 minutes

Code of Conduct

I T. Subahan (192511121) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student:



Annexure A – Agile Sprint Planning & User Story Workshop

Instructions to Students:

Each student is assigned an individual software project problem statement. Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

Based on the given problem statement, perform an **Agile Sprint Planning and User Story Workshop** by applying Agile software engineering practices. Prepare a comprehensive sprint planning document covering the following aspects:

1. **Identify the Project Vision and Stakeholders**
   - Define the project's vision, objectives, and identify all key stakeholders involved.
2. **Create Epics and User Stories**
   - Break down the requirements into Epics and formulate well-structured User Stories using the format: *As a <user>, I want <feature>, so that <benefit>.*
3. **Define Acceptance Criteria**
   - Write clear and measurable acceptance criteria for each selected user story using the Given–When–Then format or equivalent.
4. **Prioritize the Product Backlog**
   - Organize and prioritize the product backlog based on business value, customer needs, and project dependencies.
5. **Plan the Sprint Backlog**
   - Select high-priority user stories for the first sprint and divide them into actionable development tasks.
6. **Estimate Effort Using Story Points**
   - Assign story points to each selected user story using an appropriate estimation technique (e.g., Fibonacci sequence or Planning Poker) and justify your estimates.
7. **Present the Sprint Goal and Expected Deliverables**
   - Define a clear sprint goal and describe the expected sprint deliverables, including the features to be completed at the end of the sprint.

Rubrics for Calculation

Criteria	Excellent (5 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Project Vision & Stakeholder Identification (30%)	Project vision is clear, comprehensive, and aligned with the problem statement. All relevant stakeholders and their roles are correctly identified.	Vision is clear with minor omissions. Most stakeholders are correctly identified.	Vision is partially defined. Some stakeholders are missing or incorrectly identified.	Vision is unclear or incomplete. Stakeholders are poorly identified or missing.
Epics & User Stories (25%)	Epics are well-defined and user stories are complete, follow Agile format, and fully capture functional requirements.	Epics and user stories are mostly complete with minor formatting or requirement issues.	User stories are partially complete with limited Agile structure.	User stories are incomplete, unclear, or do not follow Agile practices.
Acceptance Criteria (20%)	Acceptance criteria are specific, measurable, testable, and aligned with all user stories.	Acceptance criteria are mostly clear and testable with minor gaps.	Acceptance criteria are partially defined and lack clarity or completeness.	Acceptance criteria are missing or not aligned with user stories.
Product Backlog Prioritization & Sprint Planning (15%)	Product backlog is logically prioritized, sprint backlog is realistic, and task breakdown is complete.	Backlog prioritization and sprint planning are mostly appropriate with minor issues.	Prioritization or sprint planning is partially complete with limited justification.	Backlog prioritization and sprint planning are poorly organized or incomplete.
Story Point Estimation, Sprint Goal & Presentation (10%)	Story point estimation is well justified, sprint goal is clear, deliverables are realistic, and presentation is professional and well organized.	Estimation and sprint goal are mostly appropriate with minor inconsistencies. Presentation is clear.	Estimation or sprint goal lacks justification. Presentation is adequate but incomplete.	Estimation is unrealistic or missing. Sprint goal, deliverables, and presentation are unclear or incomplete.

Criteria	Mark
Project Vision & Stakeholder Identification **(30%)**	/5
Epics & User Stories **(25%)**	/5
Acceptance Criteria **(20%)**	/5
Product Backlog Prioritization & Sprint Planning **(15%)**	/5
Story Point Estimation, Sprint Goal & Presentation **(10%)**	/5
TOTAL	/25

Vehicle Service and Maintenance Management System

1. Introduction

The Vehicle Service and Maintenance Management System is a software application designed to help vehicle owners manage their vehicle servicing activities efficiently. It allows users to book service appointments online, maintain service history, receive maintenance reminders, and track the status of their vehicles during servicing.

Traditional vehicle service management often depends on phone calls, paper records, and manual reminders. These methods can cause delays, missing records, and difficulty in tracking service progress. The proposed system automates these activities and provides a centralized platform for vehicle owners and service center administrators.

The project uses Agile project management techniques and advanced software development methodologies to improve flexibility, teamwork, software quality, and timely delivery.

2. Problem Statement

Vehicle owners often face difficulties booking service appointments, remembering maintenance schedules, accessing previous service records, and knowing the current status of their vehicles. Service centers may also struggle with managing customer details, service requests, spare parts, and technician assignments manually.

Therefore, a computerized system is required to simplify vehicle servicing, automate reminders, maintain accurate service records, and improve communication between customers and service centers.

3. Objectives

- To provide online vehicle service appointment booking.
- To maintain a digital record of vehicle service history.
- To send reminders for oil changes, inspections, and scheduled maintenance.
- To enable real-time or regularly updated service status tracking.
- To help service center staff manage bookings and assign technicians.
- To improve customer satisfaction, efficiency, and data accuracy.
- To apply Agile methods for incremental development and continuous feedback.

4. Main Modules

Module 1: User and Vehicle Management

Users register, log in, and add vehicle details such as registration number, model, make, and manufacturing year.

Module 2: Service Appointment Booking

Customers select the required service, preferred date, time slot, and service center. The system records and confirms the appointment.

Module 3: Service History Management

The system stores previous service dates, work performed, spare parts replaced, costs, and technician remarks.

Module 4: Maintenance Reminders

The system generates reminders based on service dates, time intervals, or recorded vehicle

mileage.

Module 5: Service Status Tracking
Customers can view status updates such as Booking Confirmed, Vehicle Received, In Service, Waiting for Parts, Completed, and Delivered.

5. Proposed System Architecture

User Interface
Customer Portal · Admin Dashboard · Technician Portal
Application and Business Logic
Booking · Reminders · Service Tracking · History
Database Layer
Users · Vehicles · Appointments · Service Records · Notifications
The user interface collects requests from customers and service center staff. The application layer processes appointments, service updates, and maintenance schedules. The database stores all relevant information securely for future access.

6. Advanced Software Development Methodologies

The project can apply the following methodologies to improve development and management.

Methodology	**Application in the project**
Agile development	Build the application in small, manageable increments.
Scrum framework	Organize work into short sprints with defined goals.
DevOps	Automate testing, integration, and deployment where possible.
Test-Driven Development (TDD)	Write tests before implementing selected functions.
Continuous Integration (CI)	Test new code whenever team members integrate changes.
Risk management	Identify risks such as data loss, missed reminders, and scheduling conflicts.

Recommended approach: Use Scrum as the primary project management framework, supported by CI, automated testing, and regular feedback.

7. Agile Project Management Techniques

A. Product backlog

A product backlog is a prioritized list of features and tasks required to develop the system.

Priority	**Backlog item**
High	User registration and login
High	Add and manage vehicle details
High	Book service appointments
High	Update and track service status
Medium	View service history
Medium	Schedule maintenance reminders
Medium	Admin dashboard and technician assignment
Low	Reports and service analytics

B. Sprint planning

The development work is divided into short iterations called sprints. Each sprint delivers a working part of the application.

Sprint Duration Planned deliverables

Sprint 1 Week 1–2 Requirements, UI design, login, vehicle management

Sprint 2 Week 3–4 Appointment booking and scheduling

Sprint 3 Week 5–6 Service history and status tracking

Sprint 4 Week 7–8 Reminders, testing, integration, and final deployment

C. Daily Scrum meeting

The team conducts a short daily meeting to discuss:

- What was completed yesterday?
- What will be done today?
- Are there any blockers or difficulties?

This helps team members coordinate work and resolve problems quickly.

D. Sprint review

At the end of each sprint, the team demonstrates the completed features to the project supervisor or stakeholders. Their feedback is used to improve the next sprint.

E. Sprint retrospective

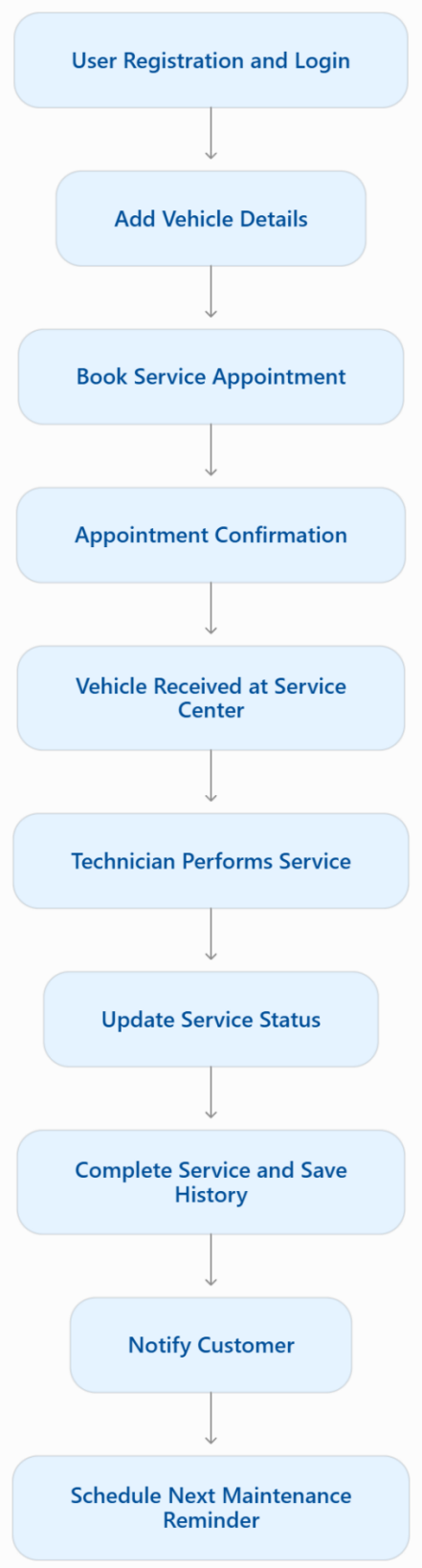
The team discusses what worked well, what problems occurred, and what changes could improve the next sprint.

8. User Stories

User stories describe system requirements from the user's perspective.

- US1: As a vehicle owner, I want to register my vehicle so that I can manage its service information.
- US2: As a customer, I want to book an appointment so that I can get my vehicle serviced at a convenient time.
- US3: As a vehicle owner, I want to view previous service records so that I can monitor maintenance history.
- US4: As a customer, I want to receive maintenance reminders so that I do not miss scheduled servicing.
- US5: As a customer, I want to track service status so that I know when my vehicle will be ready.
- US6: As an administrator, I want to assign technicians and manage bookings so that the service center operates efficiently.

9. System Workflow



10. Testing and Quality Assurance

The following testing techniques help verify that the system works correctly.

1. Unit testing: Test individual functions such as login validation and appointment creation.

2. Integration testing: Verify that booking, database storage, and notifications work together.
3. System testing: Test the complete application against its requirements.
4. Usability testing: Check whether customers can easily book appointments and track service status.
5. Security testing: Check access control, input validation, and protection of customer information.
6. Acceptance testing: Confirm with stakeholders that the application meets their needs.

11. Tools and Technologies

Purpose	Suggested tools
Project management	Jira
UI and prototype design	Figma
Frontend	HTML, CSS, JavaScript
Backend	Python Flask or Java Spring Boot
Database	MySQL
Version control	Git and GitHub
Testing	Pytest or JUnit
CI/CD	GitHub Actions

These are suggested tools; the team can choose alternatives based on course requirements and available resources.

12. Expected Outcomes

- Faster and more organized service appointment booking.
- Reduced dependence on paper-based records.
- Easy access to vehicle maintenance history.
- Fewer missed maintenance schedules through reminders.
- Better communication between customers and service staff.
- Improved project visibility through Agile tracking.
- Higher software quality through iterative testing and feedback.

13. Conclusion

The Vehicle Service and Maintenance Management System provides an efficient solution for managing vehicle servicing activities. It combines appointment booking, service history, maintenance reminders, and service status tracking in a centralized application.

By applying Agile project management, Scrum, continuous integration, and systematic testing, the development team can deliver the software incrementally, adapt to changing requirements, and improve the quality of the final product. The system benefits both vehicle owners and service centers by making maintenance activities more reliable, transparent, and convenient.